## Data Mining – Group Project

Master's in Data Science and Advanced Analytics
NOVA Information Management School
Universidade Nova de Lisboa

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# Bonus Segment

## Amazing International Airlines Inc.


**Group 40**

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## 1 Introduction

To further enhance the strategic value of the customer segmentation developed for Amazing International Airlines Inc. (AIAI), two complementary advanced analytical approaches were explored. First, a Financial Impact Model based on Customer Lifetime Value (CLV) was implemented to assess the economic relevance of each cluster. Second, a Fuzzy Clustering approach was adopted to account for ambiguity in customer behavior and overlapping segment characteristics.

## 2 Financial Impact Modeling

The objective of this component was to bridge the gap between behavioral based customer clusters and financial outcomes, by quantifying the economic value associated with each segment. This approach ensures that marketing resources are allocated to customer groups that generate the highest return.

### 2.1 CLV Integration

The CLV metric was integrated at the cluster level to assess the economic potential of the behavioral segments identified in the main analysis (Fig. 1). The figure shows the distribution of customers across low, medium and high CLV bins within each final cluster, indicating a uniform CLV composition across segments. CLV values were kept on their original scale to preserve interpretability.

A CLV score was assigned based on CLV bins and averaged per cluster. This aggregation was aligned with the final behavioral profiles, enabling economic value to be interpreted jointly with engagement, activity patterns and churn risk. Based on this combined perspective, clusters were grouped into three strategic priority levels.

**High Priority**: The Elite (Cluster 0) exhibits consistently high income, above-average activity and stable year round engagement, resulting in structurally high CLV and a high cost of customer loss. This segment represents the core value-generating group and justifies sustained investment in retention and premium services. The Ghost (Cluster 5), although currently inactive, also shows a high average CLV, likely reflecting past high activity and significant historical value. Strategically, this segment is a relevant target for win back campaigns, as reactivating former high value customers may be more cost effective than acquiring new ones.

**Medium Priority** The Autumn Traveler (Cluster 3) and The Winter Enthusiast (Cluster 4) are characterized by strong seasonal concentration and moderate activity. Their intermediate CLV aligns with the behavioral analysis, indicating customers who generate value in specific contexts but lack consistent engagement and therefore present growth potential through targeted, season-specific offers and selective bundles.

**Low Priority**: The Loyal Traveler (Cluster 1), despite the highest travel intensity and long program tenure, displays a comparatively lower average CLV, suggesting limited individual profitability due to lower margins or high benefit utilization. Given their strong loyalty and low churn risk, additional marketing investment is expected to yield diminishing marginal returns. The Casual Traveler (Cluster 2) combines low engagement, lower income and sporadic travel behavior, resulting in consistently low CLV, making cost efficient actions more appropriate than intensive retention efforts.

### 2.2 ROI Simulation and Marketing Strategy

To validate the proposed prioritization strategy, we conducted a simplified return on investment (ROI) simulation at the cluster level based on assumed marketing costs, expected uplift and targeting efficiency. This simulation was based on the following parameters:

**Marketing Costs:** Allocated at 25, 15 and 3 units per customer for High, Medium and Low priority, respectively.

**Expected Lift:** A projected increase in customer value varies from 1.5% (High) to 1.0% (Medium) and 0.3% (Low).

**Targeting Efficiency:** We assumed a more aggressive targeting reach for high value segments (60%) compared to low value ones (20%).

The simulation indicates that Low Priority clusters achieve higher ROI ratios (Fig. 2), primarily due to lower marketing costs and more conservative targeting assumptions. However, these returns are generated from customers with lower individual value, limiting the overall economic impact per customer.

In contrast, High Priority clusters show lower ROI ratios but higher customer lifetime value, meaning that even small uplifts generate substantial value gains per customer. The prioritization strategy reflects a deliberate trade off between ROI efficiency and overall value creation, favoring high value segments despite higher intervention costs.

### 2.3 Investment Timeline

From an investment perspective, the proposed strategy follows a phased implementation timeline aligned with segment priority, value potential and expected return on investment (ROI).

Short term returns (1–2 months) are driven by high priority win back opportunities, most notably Cluster 5, The Ghost. Although currently inactive, these customers exhibit high historical value and targeted reactivation efforts are expected to generate rapid uplift despite higher intervention costs, resulting in ROIs of approximately 3.9.

Early to mid term returns (3–5 months) are expected from both high and low priority segments, namely Clusters 0 and 1, The Elite and The Loyal Traveler. These clusters combine high income, frequent travel and long program tenure, leading to substantial absolute value creation. While their ROI ratios vary (3.8–6.9), their high CLV makes them strategically important contributors to overall profitability. The early inclusion of low priority segments reflects their higher short term ROI efficiency, whereas medium priority segments are not prioritized in early phases due to the seasonal concentration of their value generation, which postpones return realization.

Medium term returns (6–8 months) are associated with Cluster 2, The Casual Traveler, a low priority segment characterized by low engagement and seasonal travel behavior. Targeted incentive based offers for this segment are expected to yield ROIs in the range of 6.3.

Long term returns (9–12 months) are driven by seasonality-focused segments, specifically Clusters 3 and 4, The Autumn Traveler and The Winter Enthusiast. These clusters generate medium-priority ROI levels (4.1–4.3) primarily through niche, season specific product bundles rather than sustained volume growth.

Overall, the phased investment strategy balances high absolute value creation in high priority segments with high ROI efficiency in lower priority clusters.

## 3 Fuzzy Clustering Implementation

Traditional clustering (K-Means) assigns each customer to exactly one group. However, in the airline industry, customer behavior is often fluid. We implemented **Fuzzy C-Means (FCM)** to allow for "soft" memberships, where a customer can belong to multiple clusters with varying degrees of membership (Fig. 4).

This analysis builds directly on the hard clustering results presented in the main report, using the same feature spaces (VE and PP) to assess the robustness and interpretability of the original segmentation.

### 3.1 Fuzzy Partition Coefficient (FPC) Analysis

The FPC measures how clear the partition is (closer to 1 indicates clearer cluster separation). We applied this to our two main perspectives:

| **Perspective** | **FPC** | **High-Confidence Customers (≥0.7)** |
| --- | --- | --- |
| Value Efficiency (VE) | 0.4070 | ∼20% |
| Product Preference (PP) | 0.3363 | ∼9% |

Table 1: Fuzzy Clustering Results for VE and PP Perspectives

These values confirm that the VE segmentation identified in the main analysis is structurally more stable than PP, while still exhibiting a non negligible degree of overlap that is masked by hard assignments.

### 3.2 Strategic Insights from Ambiguity

The Fuzzy Partition Coefficient (FPC) and membership degree distributions reveal substantial overlap between customer segments (Fig. 3), particularly in the Patterns & Preferences (PP) perspective. The lower FPC observed in PP (0.336) indicates that customer preferences are not organized into sharply separated pattern based clusters. Instead, many customers exhibit mixed behavioral patterns, alternating between, for example, Economy and Business travel depending on context, trip purpose or price sensitivity.

This ambiguity is further supported by the low proportion of high confidence customers (membership degree $\geq 0.7$), especially in PP (Fig. 4), where fewer than 10% of customers can be confidently assigned to a single segment. In contrast, the Value Efficiency (VE) perspective, while still exhibiting overlap, shows a higher share of high confidence memberships (20%), suggesting relatively clearer segmentation along value related dimensions.

Fig. 5 visually reinforces these findings by mapping customers in the PCA space and coloring observations by their maximum membership degree. In the VE perspective, several regions of the PCA space display relatively high membership confidence, indicating that value-related attributes generate more clearly separable customer profiles. In contrast, the PP perspective exhibits widespread low membership values across most of the PCA space, with only a narrow subset of customers showing strong affiliation to a single cluster. The pronounced concentration of high-confidence points along specific PCA directions in PP suggests that only extreme or highly specialized patterns and preferences behaviors form well-defined segments, while the majority of customers occupy overlapping regions characterized by ambiguous and context-dependent preferences.

While the K-Means clustering presented in the main report enforces clear cut segment boundaries, the fuzzy analysis reveals that a substantial share of customers lies near these boundaries, particularly in the PP perspective. This explains why several segments exhibited similar behavioral profiles despite being assigned to distinct clusters.

Compared to hard clustering approaches such as K-Means, which impose a single cluster assignment per customer, the fuzzy framework explicitly captures this uncertainty. Customers with small membership margins between their top two clusters represent “in between” profiles, for whom rigid segment based targeting may be ineffective or even counterproductive. This is particularly relevant for segments identified as medium priority in the financial analysis, where misclassification may lead to suboptimal budget allocation without generating proportional returns.

From a business point of view, this uncertainty is actionable rather than problematic. For AIAI, it suggests that hybrid or flexible offers, rather than strictly segment specific campaigns, are more appropriate for customers located at cluster boundaries. Incorporating membership degrees into targeting strategies reduces the risk of misclassification and enables more nuanced personalization, particularly in segments characterized by fluid patterns and preferences.

Accordingly, the insights from fuzzy clustering complement the prioritization framework developed in the main report by suggesting where strict segment based targeting is appropriate (high confidence VE segments) and where adaptive or hybrid strategies are more effective (ambiguous PP segments).

## A Appendix

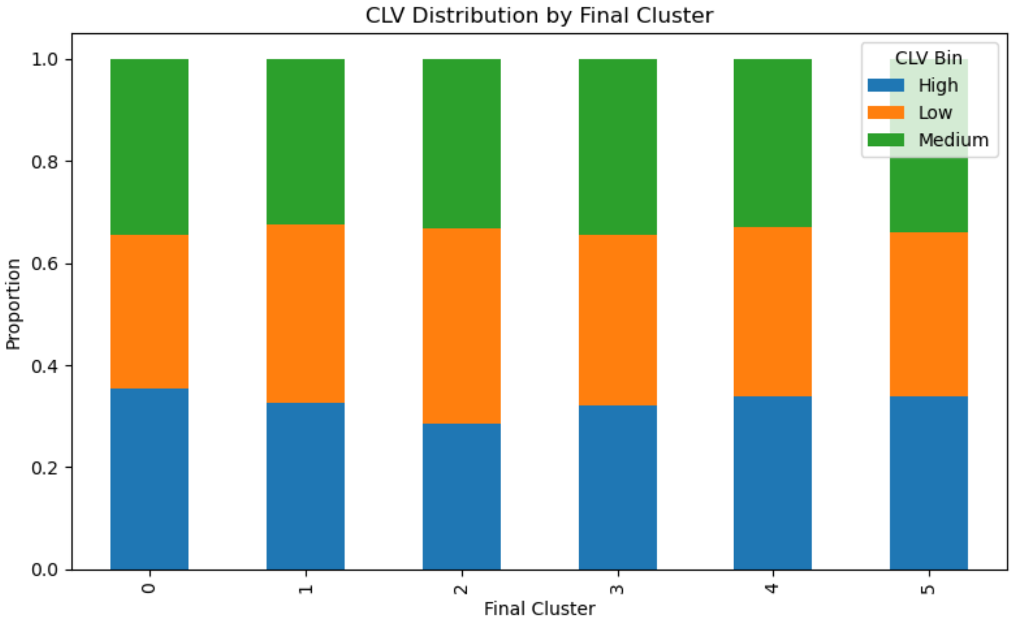


Figure 1: CLV Distribution per Final Cluster

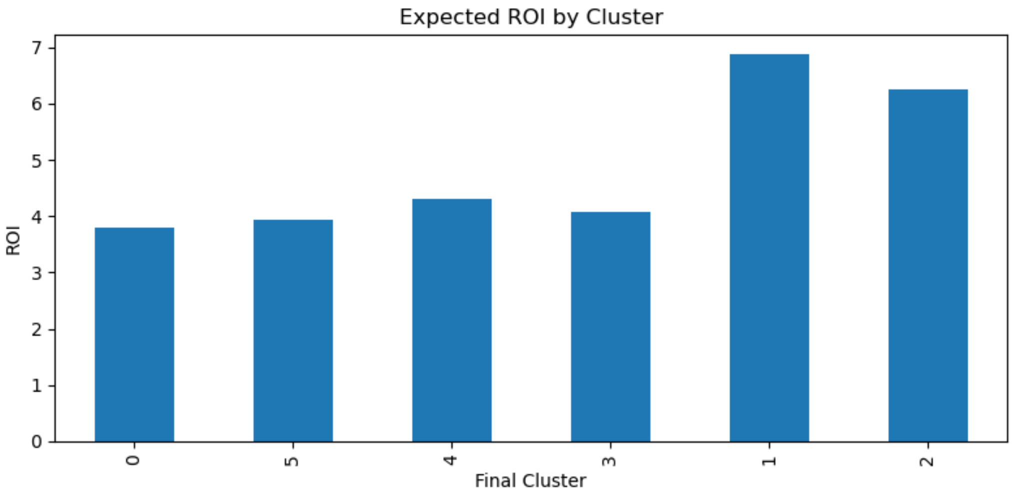


Figure 2: Projected ROI per Cluster

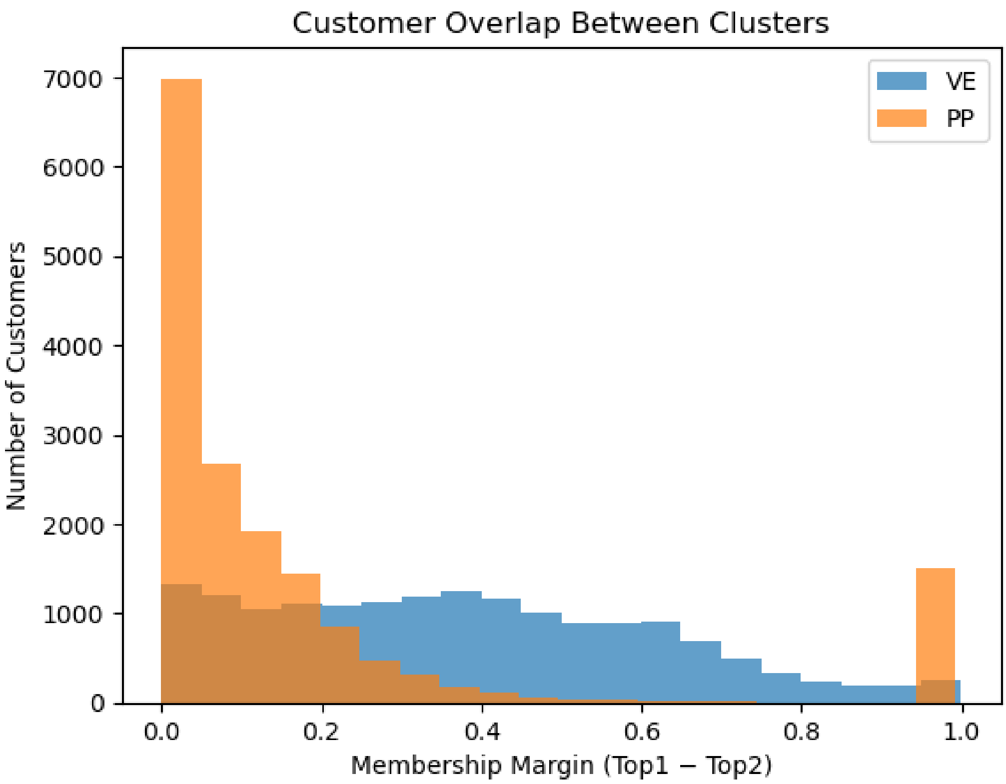


Figure 3: Customer Overlap Between Clusters

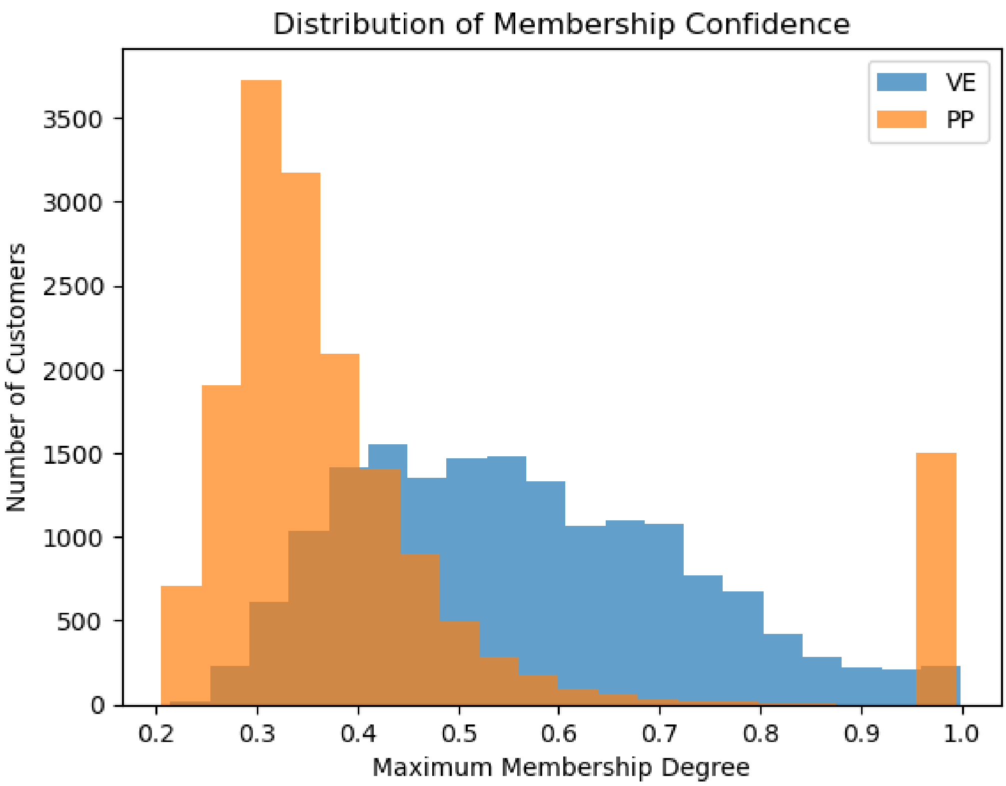


Figure 4: Distribution of Membership Confidence

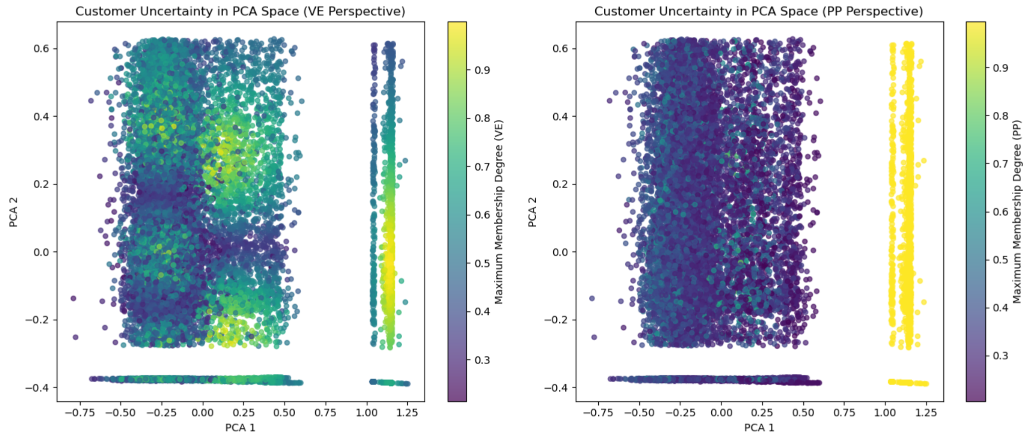


Figure 5: Customer Uncertainty in PCA Space